

Multi-Dimensional LSTM Networks for Video Prediction

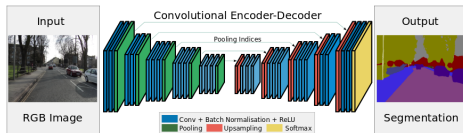
Wonmin Byeon

NVIDIA Research

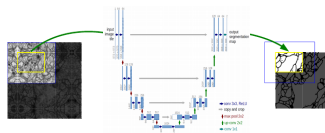
March 29, 2018



Convolutional Neural Networks



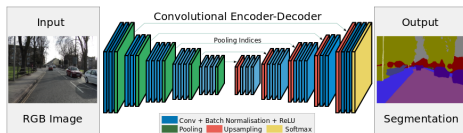
Semantic Segmentation
SegNet [Badrinarayanan16]



Biomedical Image Segmentation
U-Net [Ronneberger15]

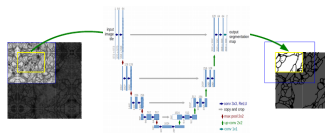
- Needs a lot of computations

Convolutional Neural Networks



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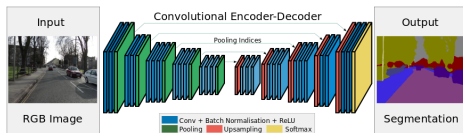


Biomedical Image Segmentation

U-Net [Ronneberger15]

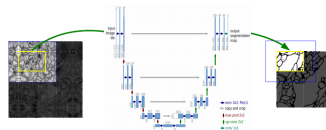
- Needs a lot of computations
 - Each window can be computed in parallel
 - An efficient GPU implementation is possible

Convolutional Neural Networks



Semantic Segmentation

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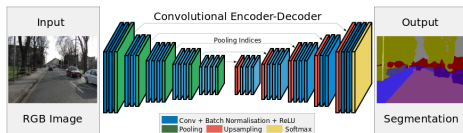


Biomedical Image Segmentation

U-Net [Ronneberger15]

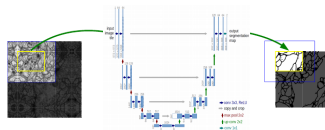
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- Has a fixed size of receptive field

Convolutional Neural Networks



Semantic Segmentation

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Biomedical Image Segmentation

U-Net [Ronneberger15]

- Needs a lot of computations
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Convolutional Neural Networks



Image



FCN Results



Ground truth

- Needs a lot of computations
 - Each window can be computed in parallel
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- Perceives only small local contexts of the pixels

Convolutional Neural Networks



Image



FCN Results



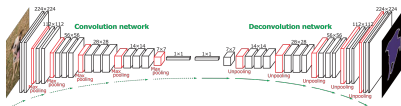
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Solutions?

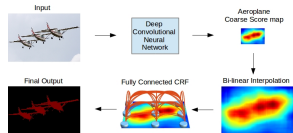
Convolutional Neural Networks: solutions

Up-pooling (deconvolution)



DeconvNet [Noh16]

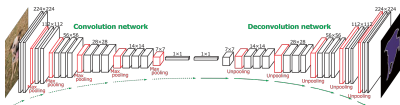
Adding Conditional Random Field (CRF)



DeepLab [Chen16]

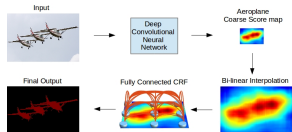
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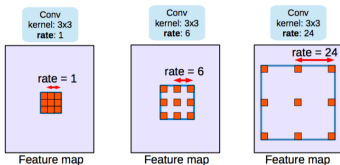
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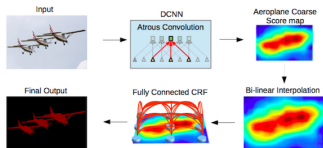


DeepLab [Chen16]

Using Dilated/Atrous Convolutions



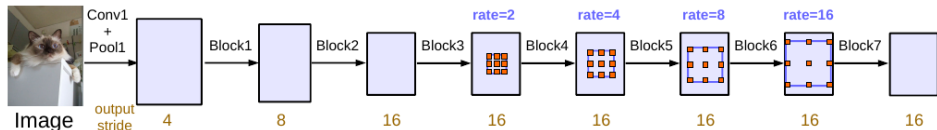
Dilated Convolutions [Yu15]



DeepLab V2 [Chen16]

Convolutional Neural Networks: solutions?

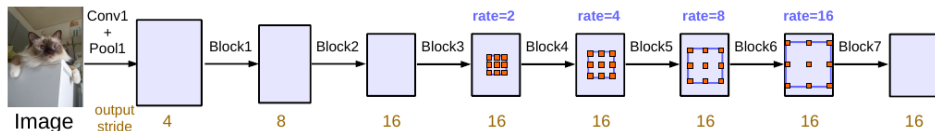
Using Dilated Convolutions & Going Deeper



DeepLab V3 [Chen17]

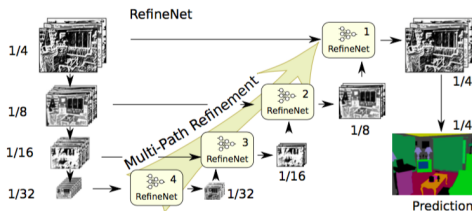
Convolutional Neural Networks: solutions?

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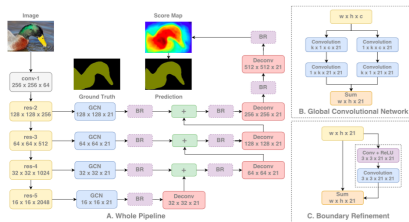
DeepLab V3 [Chen17]

Fusing Multi-Resolutions



RefineNet [Lin16]

Adopting Large Kernels



Global-CNN [Peng17]

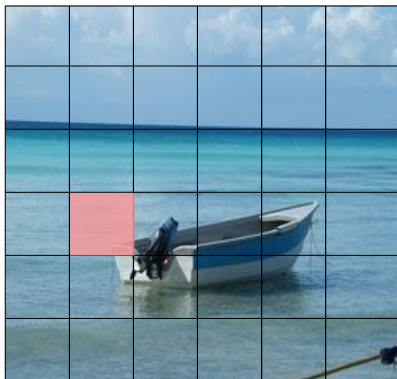
How can we efficiently capture global/long range context?



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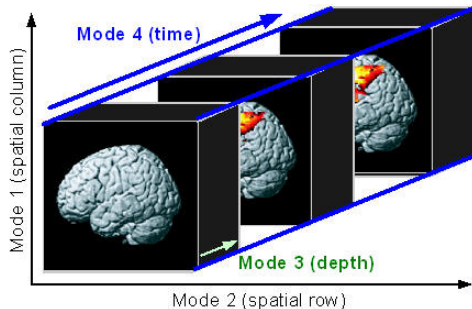


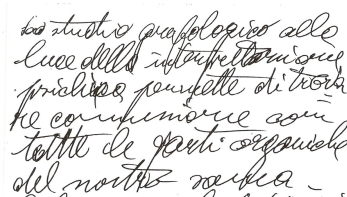
Image from <http://staffwww.dcs.shef.ac.uk/people/H.Lu/feeler.html>

Long Short Term Memory Recurrent Networks

LSTM Networks for Sequence Learning



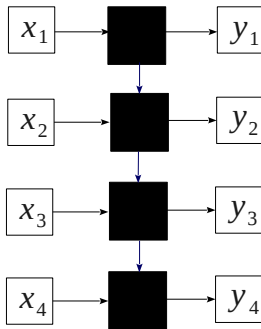
speech [Graves05, Graves06]



handwriting [Liwicki07, Graves09]

Sequence Classification Task with Dependencies

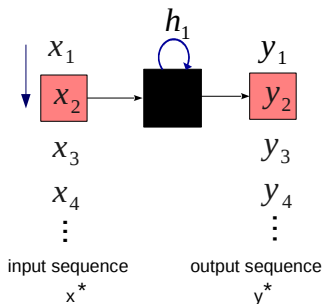
mapping $x_1x_2\dots$ to $y_1y_2\dots$



$$F: x \in X^* \rightarrow y \in Y^*$$

Sequence Classification Task with Dependencies

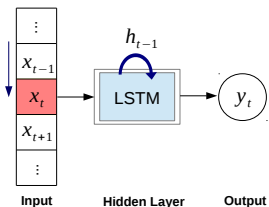
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1-Dimensional LSTM Networks

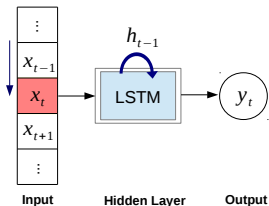
Standard LSTM

[Hochreiter97,Gers99]

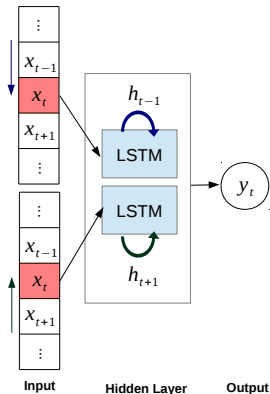


1-Dimensional LSTM Networks

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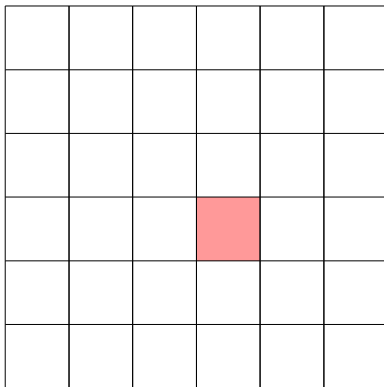
Bidirectional LSTM [Graves05, Chen05]



Multi-Dimensional LSTM networks

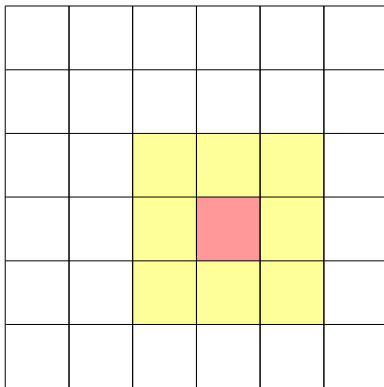
2-Dimensional LSTM Networks for images

red: the current pixel



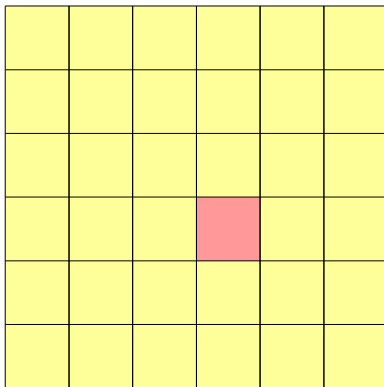
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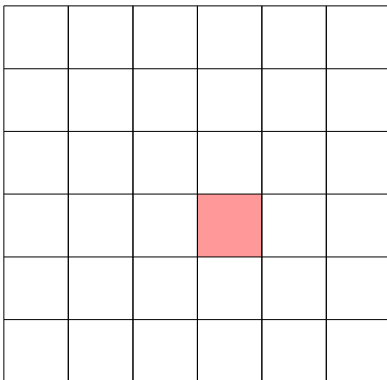
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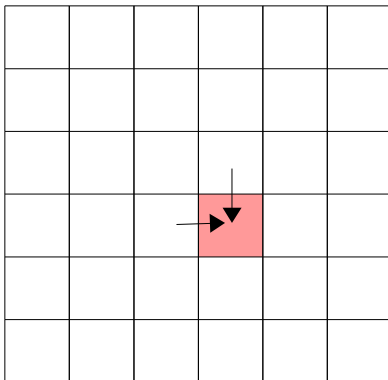
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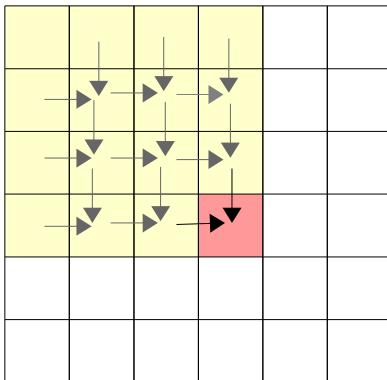
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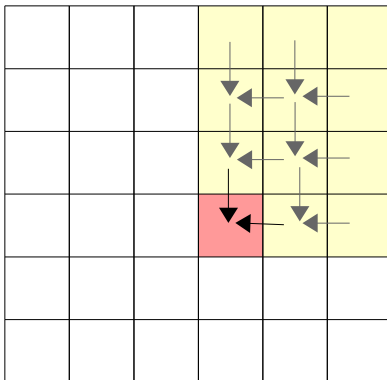
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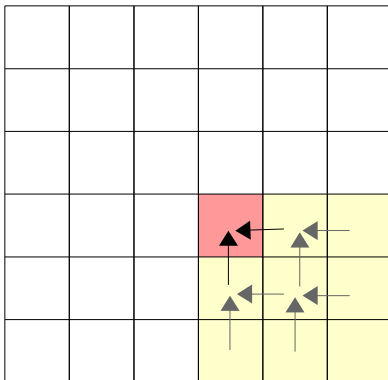
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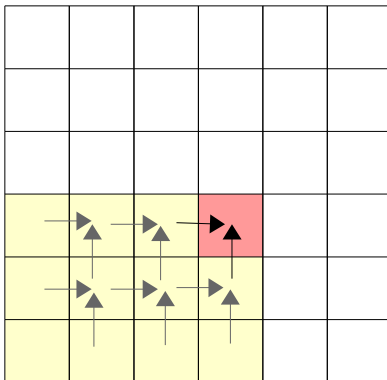
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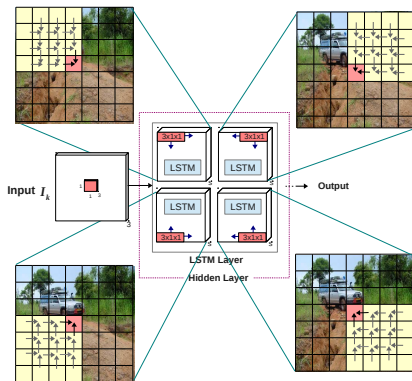


2-Dimensional LSTM Networks for images

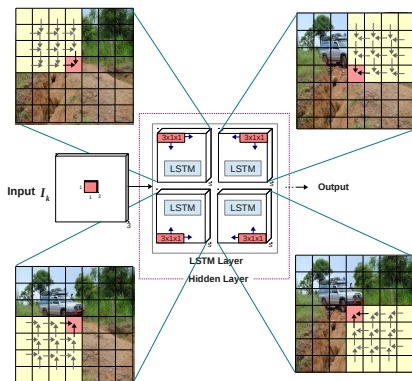
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2-Dimensional LSTM Networks for images

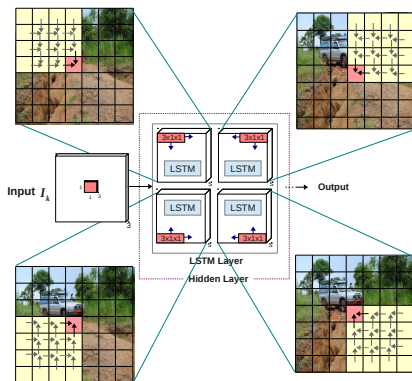


2-Dimensional LSTM Networks for images



- Perceives **the entire spatio-temporal context** of each pixel in a few sweeps through all pixels
- Requires **fewer number of parameters** to takes both local and global contexts into account
- **End-to-End learning**, No pre- and post- processing

2-Dimensional LSTM Networks for images



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- Requires **fewer number of parameters** to takes both local and global contexts into account
- **End-to-End learning**, No pre- and post- processing
- Needs a lot of computations
- Is **hard to parallelize** due to the dependencies between pixels

**Too slow.. Can we use the model
for more than 2-Dimensional data?**

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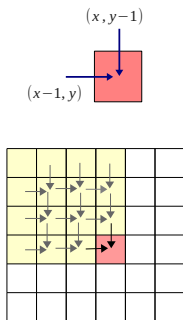


Parallel Computing?

Parallel Multi-Dimensional LSTM

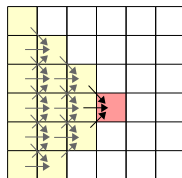
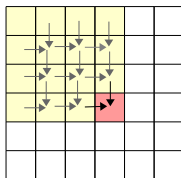
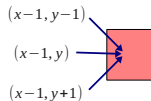
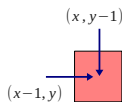
Parallel Multi-Dimensional LSTM, With Application to Fast Biomedical Volumetric Image Segmentation, [Stollenga&Byeon15]

Parallel MD-LSTM Networks



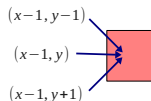
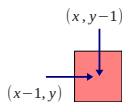
MD-LSTM

Parallel MD-LSTM Networks

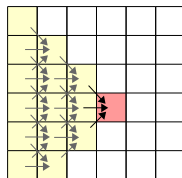
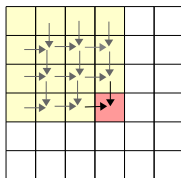


MD-LSTM Parallel MD-LSTM

Parallel MD-LSTM Networks

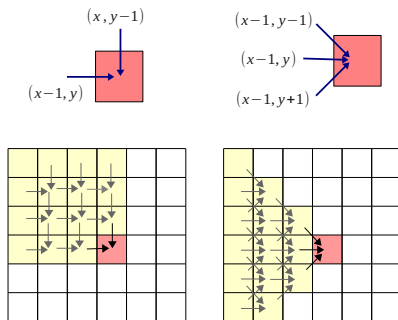


- Retain all advantages of LSTMs



MD-LSTM Parallel MD-LSTM

Parallel MD-LSTM Networks

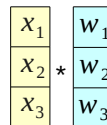
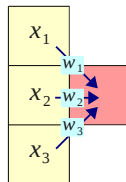
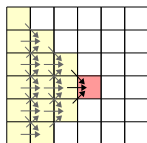


MD-LSTM Parallel MD-LSTM

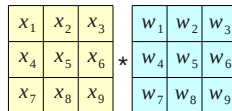
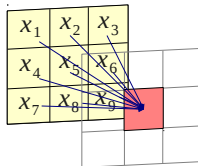
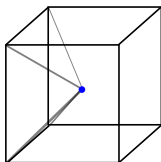
- Retain all advantages of LSTMs
- Easy to parallelize on GPUs
 - All operations can be implemented using convolutions

Parallel MD-LSTM Networks

2D images

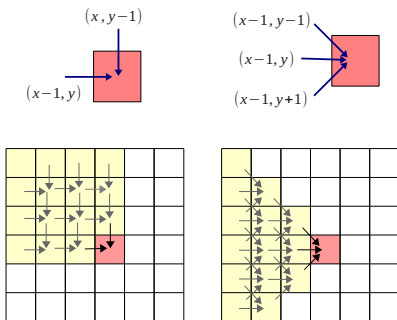


3D volumetric images



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Parallel MD-LSTM Networks



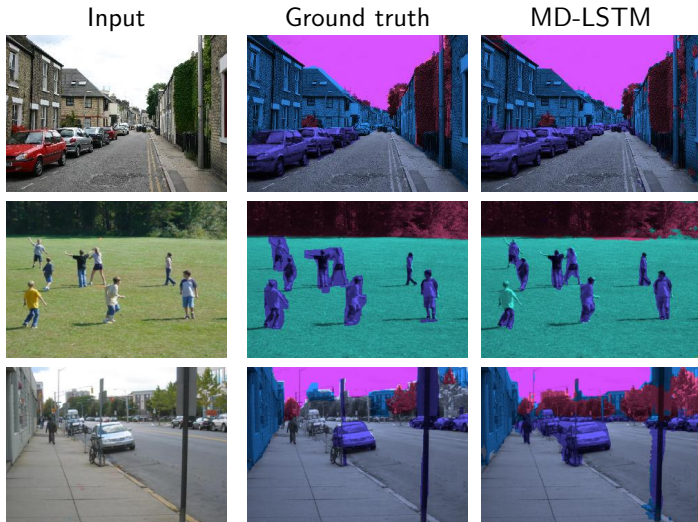
MD-LSTM

Parallel MD-LSTM

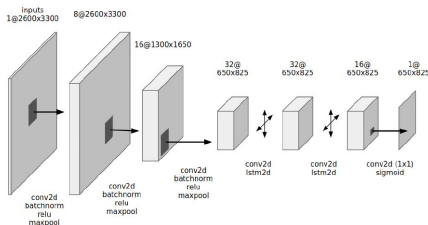
- Retain all advantages of LSTMs
- Easy to parallelize on GPUs
 - All operations can be implemented using convolutions
- More efficiently covers volume
 - PyraMiD-LSTM: 6 pyramids

Applications

Application 1: Image Segmentation



Application 2: Layout Analysis



- Simultaneously performs: text/image segmentation, border noise removal, text/math segmentation, text line detection
- 1 sec / page (GPU)

Training for Different Detection Tasks



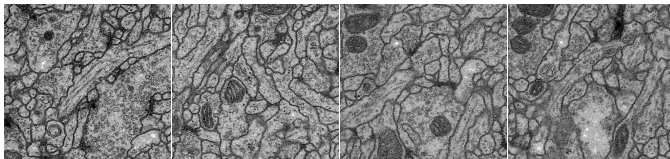
original page body text detection text detection text frame detection border noise detection

Application 3: Medical Image Segmentation

Membrane Segmentation on Electron Microscopy Images

Rand error - human: 0.002, CNNs [Ciresan12]: 0.050, MD-LSTM: **0.047**

Input



CNNs



(after post-processing)

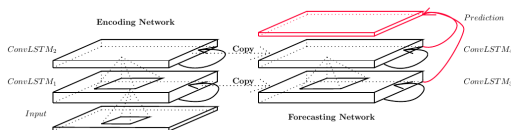
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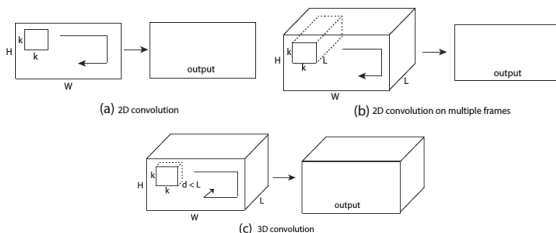
Video Prediction

Video Prediction



Convolutional LSTM (ConvLSTM)

[Shi15, Finn16, Lotter16, Villegas17]

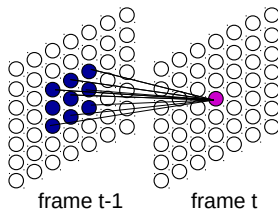


t : time frame

3D CNN

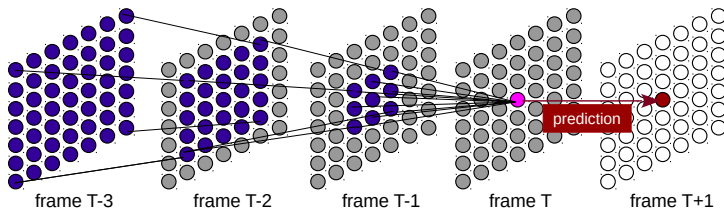
[Tran15, Mathieu15]

Convolutional LSTM for Video Prediction



Convolutional LSTM for Video Prediction

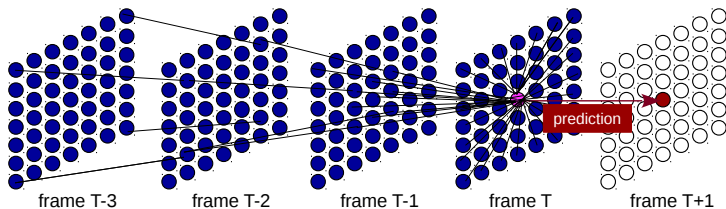
Blind Spot Problem



red: current pixel **blue:** context covered **gray:** blind spots

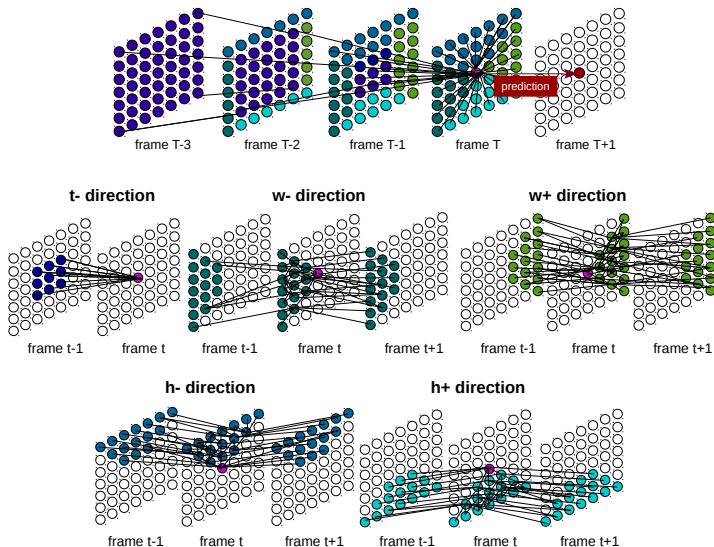
Convolutional LSTM for Video Prediction

Solution?

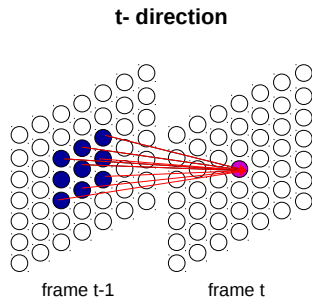


red: current pixel **blue:** context covered **gray:** blind spots

Fully Context-Aware Video Prediction



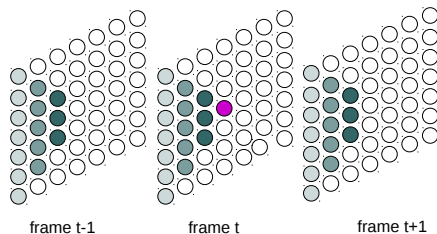
Fully Context-Aware Video Prediction



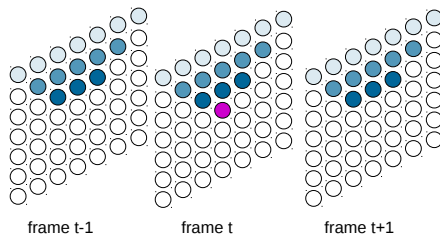
Fully Context-Aware Video Prediction

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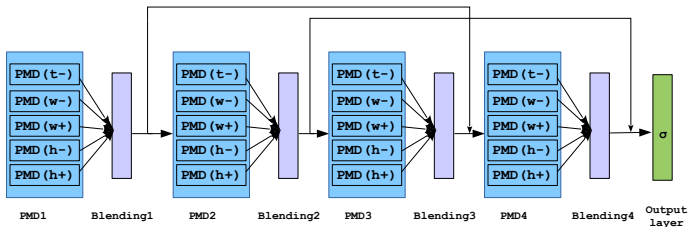
w- direction



h- direction



Fully Context-Aware Video Prediction: ContextVP



- Parallel MD-LSTM (PMD) Block: s^d , $d \in D = \{h-, h+, w-, w+, t-\}$
- Blending Block
 - Uniform Blending: $m = f((\sum_{d \in D} s^d) \cdot W + b)$
 - Weighted Blending:

$$m = f(S \cdot W + b), \quad S = [s^{t-} \quad s^{h-} \quad s^{h+} \quad s^{w-} \quad s^{w+}]^T$$
- Directional Weight Sharing (DWS)
- Loss: $L1 + \text{Gradient Difference Loss (GDL)}$ [Mathieu15]

Fully Context-Aware Video Prediction: Baseline

- A stack of 20 ConvLSTM layers + 2 skip connections
- The number of parameters comparable to our best model

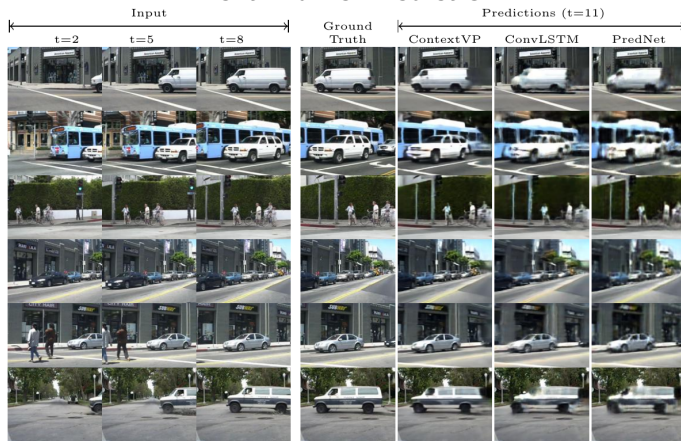
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- **It outperforms almost all state of the art models!**

Fully Context-Aware Video Prediction: Baseline

- A stack of 20 ConvLSTM layers + 2 skip connections
- The number of parameters comparable to our best model
- **It outperforms almost all state of the art models!**
- Compare to ContextVP?
 - ContextVP **performs better** than the baseline with the similar number of parameters
 - ContextVP is more suitable for **parallelization**: pixel-level and direction-level parallelization are possible

Next-Frame Prediction



Train: KITTI dataset, Test: CalTech Pedestrian dataset, 10 input frames

- ContextVP: 4 layers with Weighted Blending and DWS
- ConvLSTM: 20 ConvLSTM layers (baseline)
- PredNet: Deep Predictive Coding Network [Lotter16]

ContextVP: Fully Context-Aware Video Prediction [Byeon17]

Multi-Step Prediction

- Trained for next-frame prediction
- Tested to recursively predict 8 future frames

UCF-101 dataset, 4 input frames

- ContextVP: 4 layers with Weighted Blending and DWS
- ConvLSTM: 20 ConvLSTM layers (baseline)
- MCnet [Villegas17]

ContextVP: Fully Context-Aware Video Prediction [Byeon17]

Multi-Step Prediction

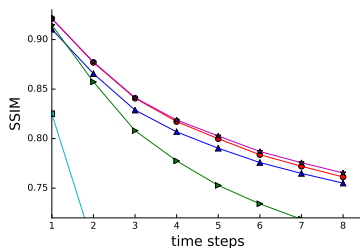
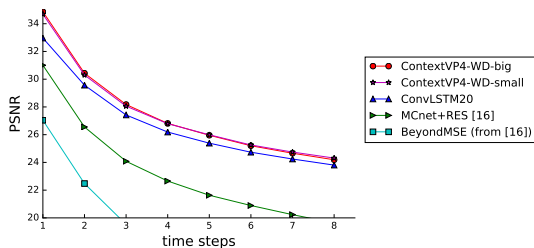
- Trained for next-frame prediction
- Tested to recursively predict 8 future frames

UCF-101 dataset, 4 input frames

- ContextVP: 4 layers with Weighted Blending and DWS
- ConvLSTM: 20 ConvLSTM layers (baseline)
- MCnet [Villegas17]

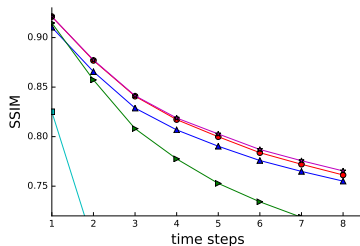
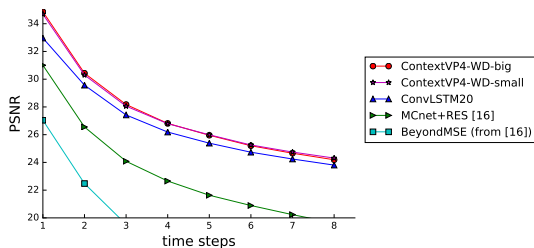
ContextVP: Fully Context-Aware Video Prediction [Byeon17]

Multi-Step Prediction

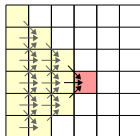
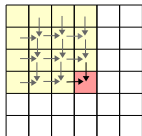
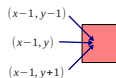


- ContextVP4-WD-small: 2.0M parameters
- ContextVP4-WD-big: 8.6M parameters
- ConvLSTM20: 9.0M parameters
- MCnet: 14M parameters
- BeyondMSE: 8.9M parameters

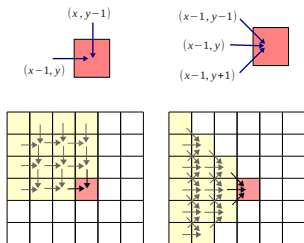
Multi-Step Prediction



- ContextVP4-WD-small: 2.0M parameters
- **ContextVP4-WD-big: 8.6M parameters**
- ConvLSTM20: 9.0M parameters
- MCnet: 14M parameters
- BeyondMSE: 8.9M parameters



MD-LSTM Parallel MD-LSTM



MD-LSTM Parallel MD-LSTM

Applications



image segmentation

[Byeon15]

body text
detection

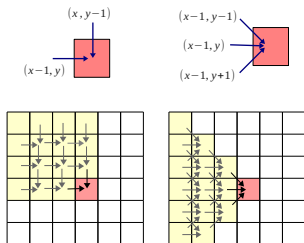
text detection

layout analysis

[Breuel17]

medical image
segmentation

[Stollenga&Byeon15]



MD-LSTM Parallel MD-LSTM

Applications



image segmentation

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body text
detection

text detection

layout analysis

[Breuel17]

medical image
segmentation

[Stollenga&Byeon15]

Fully Context-Aware Video Prediction [Byeon17]

Paper: <https://arxiv.org/abs/1710.08518>

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Thank you

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